Knowledge Representation – Project Report

Healthcare Clinical Ontology

A Formal OWL 2.0 Ontology for Hospital Clinical Environments

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1. Introduction and Objective

This report documents the design, implementation, and validation of the Healthcare Clinical Ontology, developed in fulfilment of the exam proposal assigned by Prof. Matteo Cristani. The ontology formally represents the domain of hospital clinical management, using the Web Ontology Language (OWL 2.0) and the Protégé editor, in line with the admissible exam topics of "Academic Organization Ontology," "Healthcare Clinical Ontology," and related categories.

The objective of the project is to build a logically consistent, reasoner-verifiable knowledge model of a hospital environment, centred on five core entities: Patients, Doctors (and other Healthcare Providers), Diseases, Medical Records, and Treatments. The ontology is intended to support automated reasoning tasks such as classification, consistency checking, and query answering over clinical data.

2. Methodology

The ontology was developed following a standard knowledge-engineering workflow, structured into five phases:

- Domain scoping — the hospital environment was bounded to the five entity types specified in the assignment.
- Class elicitation — a class hierarchy was derived through top-down decomposition, starting from a general Person concept.
- Property definition — Object Properties were introduced to connect classes (e.g. hasDisease, treats), and Data Properties to describe their attributes (e.g. patientID, dateOfBirth).
- Formalization — the model was encoded in OWL 2.0, using the OWL/XML serialization, editable and verifiable in Protégé.
- Population and validation — representative Named Individuals were added and the ontology was checked for logical consistency using Protégé's built-in HermiT reasoner.

The resulting artifact is a single OWL/XML file, HealthcareClinicalOntology.owl, deliverable via GitHub as requested, and openable directly in Protégé for inspection or extension.

3. Class Hierarchy and Architecture

The ontology is organized around a top-level distinction between the people involved in care and the clinical entities surrounding that care. The complete hierarchy is as follows:

```
Thing
├── Person
│   ├── Patient
│   └── HealthcareProvider
│       ├── Doctor
│       │   ├── GeneralPractitioner
│       │   └── Specialist
│       └── Nurse
├── Disease
```

```
│    ├── ChronicDisease
│    └── InfectiousDisease
├── MedicalRecord
├── Treatment
│    ├── MedicalPrescription
│    ├── Surgery
│    └── Therapy
└── Medication
```

Figure 1 — Class hierarchy of the Healthcare Clinical Ontology

3.1 Design Decisions

- Patient and HealthcareProvider are declared disjoint classes, so the reasoner flags any individual mistakenly asserted as both — a common modelling error in role-based domains.
- treats and isTreatedBy are declared as inverse object properties, enabling bidirectional navigation between doctors and patients without redundant assertions.
- hasMedicalRecord and patientID are declared functional, reflecting the real-world constraint that a patient has exactly one active medical record and one identifier.

4. Object and Data Properties

4.1 Object Properties

Property	Domain	Range
hasDisease	Patient	Disease
treats / isTreatedBy	Doctor / Patient	Patient / Doctor
prescribesMedication	Doctor	Medication
hasMedicalRecord	Patient	MedicalRecord
undergoesTreatment	Patient	Treatment
recordsDisease	MedicalRecord	Disease
includesTreatment	MedicalRecord	Treatment

4.2 Data Properties

Property	Domain	Range
patientID	Patient	xsd:string (functional)
dateOfBirth	Person	xsd:date
diseaseName	Disease	xsd:string
licenseNumber	Doctor	xsd:string
dosage	Medication	xsd:string
prescriptionDate	MedicalPrescription	xsd:date

5. Population: Named Individuals

To demonstrate the ontology in operation, a set of representative individuals was asserted, together with their class membership and the object/data property relations connecting them.

Individual	Type	Key Assertions
JohnDoe	Patient	hasDisease Diabetes; undergoesTreatment InsulinTherapyRx; patientID PT-1001
MariaRossi	Patient	hasDisease Hypertension; patientID PT-1002
LucaBianchi	Patient	hasDisease Influenza; patientID PT-1003
DrSmith	Doctor	treats JohnDoe, MariaRossi; licenseNumber MD-5521
DrVerdi	Doctor	treats LucaBianchi; licenseNumber MD-7734

All patients are additionally asserted as pairwise DifferentIndividuals (and likewise the two doctors), since OWL adopts the Open World Assumption and does not infer inequality between distinct individuals by default.

6. Validation

The ontology was loaded into Protégé and checked with the HermiT reasoner (Reasoner → HermiT → Start reasoner). No unsatisfiable classes were reported, and the inferred class hierarchy matched the asserted hierarchy, confirming logical consistency. The Individuals tab was used to verify that all object and data property assertions resolved correctly against the declared domains and ranges.

7. Repository and Deliverables

The full deliverable is available as a GitHub repository containing:

- HealthcareClinicalOntology.owl — the complete ontology in OWL 2.0 / OWL-XML format
- README.md — project documentation, including setup instructions for opening the file in Protégé

The repository link will be shared separately via email, per the submission instructions for this exam proposal.

8. Conclusion

This project delivers a compact but logically rigorous OWL 2.0 ontology for the hospital clinical domain, covering the five required entity types, a disjoint and inverse-aware property structure, and a populated individual set that demonstrates correct usage. The model is verified consistent under the HermiT reasoner and is structured to allow straightforward extension — for example, adding Nurse-specific properties, Insurance entities, or SWRL rules for clinical decision support in future iterations.